Which Model Changes Can Counterfactual Explanations See? A Ground-Truth Benchmark

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Abstract. Counterfactual methods are used to explain how a classifier changed after an update, but without ground truth one cannot tell which method sees which kind of change. We contribute a benchmark with exact ground truth (rendered images, injected changes, a grid over the rotation and offset of the decision boundary) and a linear analysis of what counterfactual differencing, gradient differences, importance differences, disagreement search and differential counterfactuals (Δ-CE, minimal inputs on which exactly one model changes its decision by a margin) can see. The analysis predicts that gradient differences see a rotation and are blind to an offset; both halves were confirmed on CNNs in tests registered before the grid was run (Recovery 0.08 on offsets; +0.57 from the smallest to the largest rotation), and with quantities taken at the query it also predicts which witnesses show the new dependence. A pre-registered hypothesis that Δ-CE recovers the changed factor better than gradient differences was rejected. Margin witnesses abstain on null updates, also on a tabular dataset; at zero null false positives a thresholded differencing score detects no offset below $|b| = 2$, while witnesses detect constant logit offsets from $|b| = 1.1$ and fine-tuned offsets on every lowered-class query in 6 of 7 pairs (descriptive).

1 Introduction

Deployed classifiers are updated routinely, and an update that improves accuracy can still change individual decisions in unwanted ways [11]; explaining *how* they differ is a goal of explainable AI under distributional change [3, 12]. Artelt et al. [1, 2] compare a counterfactual [8] of the same input under each model, or approximate this by a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which two models disagree. These families assume different things about *what* changed, and without ground truth this goes unchecked.

We ask which kind of change each family can see, using injected changes with exact ground truth. We also include *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search [5, 6] at a query; unlike robust counterfactuals [9, 10], which seek a point valid for both models, it seeks one valid for exactly one, and the margin makes it return nothing for models that differ only by jitter.

Contributions. (i) A benchmark with exact ground truth for explaining model updates: five rendered factors, three injected changes, null updates, and a 4×4 grid over the rotation and threshold of the decision boundary (32 of the 49

model pairs), with code and protocol. (ii) A linear analysis separating rotation from offset, with a prediction for each family. (iii) A test of these predictions on CNNs under a protocol fixed in advance. The analysis transfers with quantities taken at the query, and its projection equivalence only with the old normal taken at the witness; Table 3 separates tests that follow from a lemma or definition from risky predictions, and all failures are reported.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness.

Linear case. For linear margins on $\mathbb{R}^p$ with independent w_o, w_n the nearest witness is unique: the ordinary counterfactual of f_{new} or the minimum-norm point with both constraints active. Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing is $u_n - u_o$, the difference of the two models' minimum-norm counterfactuals, the witness is $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference has components in ratio $\tan(\theta/2) < 1$ along $(\hat{w}_o, v)$, i.e. mainly along the new dependence v; the witness has $|u^\star \cdot \hat{w}_o| = \min\{d_n \cos\theta, d_o\}$ and is dominated by v iff $\theta > \pi/4$ or $d_o(\sin\theta + \cos\theta) < d_n$. After projecting out $\hat{w}_o$ all three coincide up to scale (in the spirit of Theorem 1 of [1], which expresses the cosine of two weight vectors through their counterfactuals).

On CNNs the last statement holds approximately only with $\hat{w}_o$ taken at the witness (Section 4.2). In words, gradient differences see a *rotation* and are blind to an *offset*; type-N witnesses exist only on the side whose margin the offset lowers; for a constant logit offset such as the prior correction [13], they abstain exactly when the shift is below 2κ (Corollary 1 in the proofs). The analysis was formulated after the main experiment and before the grid was run.

Table 1: Injected changes; $k^\star$ is the ground-truth factor (none for the grid cell $(0, 0.5)$). [a]Two pairs have no $k^\star$ ($t = 0.5$) and two failed the offset check ($t = 0.53$), so four enter H5a.

Change	Adaptation data	$k^\star$	Pairs
Shortcut	colour correlated with label (0.99)	colour	3
Subgroup	labels flipped for size > 0.75	size	3
Threshold	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid, $\alpha = 0$	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$, $t \in \{.5, .53, .56, .6\}$	aspect	8[a]
Grid, $\alpha > 0$	same, $\alpha \in \{.1, .25, .5\}$	colour	24

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0, 1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise; the renderer is exact, so the factor change of any explanation is read without a learned oracle. f_{old} is a small CNN (five seeds; two on the grid); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1). A pair enters only if its change is visible in decisions on validation data (null pairs: decisive disagreement ≤ 0.02). For the grid we measure the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} by a logistic fit to its decisions, and analyse at these measured values; all 32 realised the intended rotation, while offsets below about 0.05 were not produced reliably.

Metrics and methods. Per pair: $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0; $k^\star$ in Table 1). Recovery uses type-N witnesses only (pre-declared); a query whose class the update raises (Threshold, class 0) has none and counts 0. All counterfactual methods share one solver (Adam with a decaying penalty, then SLSQP), the same points and the same budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [1]; **Grad-diff**, the gradient difference [1]; **Imp-diff**, a permutation-importance difference [4]; **DIG**, a DIG-style line search along segments to training points, after [5]; **No-prox**, Δ-CE without the proximity term (the DeepXplore objective [6]); and **CF-new**, the ordinary counterfactual of f_{new} alone.

Protocol. Every test split served at most one confirmatory test, after all choices were fixed: split A is the locked split of Table 2, split B the new data of C1–C3, split C the fresh data of the type-O analysis. Table 3 lists every test; the grid's manipulation check, exclusion rule and statistics were fixed after a pilot and before its test data; one analysis-code error was later corrected. Bootstraps are paired and stratified by pair; they resample points within three pairs (two seeds of f_{old} on the grid), so intervals do not cover model-to-model variation, which we did not measure beyond re-solves of the same pair. The rest is descriptive. A digit tier and a label-shift update failed qualification and were stopped.

Table 2: Recovery@r (found-rate) with type-N Δ-CE, split A (locked), mean over three pairs (95% CIs within ±0.035); last row: found-rate on the 8 null pairs.

	Δ-CE	CF-diff	Grad-diff	Imp-diff	DIG	No-prox	CF-new
Shortcut	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.994	1.0	1.0	0.000	0.000	0.996

Table 3: Tests (95% CIs; †90%). K: D = follows from a lemma, proposition or definition, R = risky prediction. Formed before any result: H2a, P-tie; after the first: C1–C3; before the grid: H5a–H5d; later (new data): H6–H8. Holm: {C1, C3}, {H5a–H5d}, {H7c, H8}. Proj.: $\hat{w}_o$ projected out.

Claim	Data	Estimate	p_{Holm}	Status	K	ID
Δ-CE > Grad-diff	Subgroup	−0.09 [−0.13, −0.05]		rejected	R	H2a
Tie ($\lvert\Delta\rvert \leq 0.1$)	Shortc., Thresh.	$\lvert\Delta\rvert = 0.17$–$0.53$		not conf.	R	P-tie
Δ-CE > Grad-diff	Threshold, B	+0.31 [0.27, 0.35]	$2{\cdot}10^{-4}$	confirmed	D	C1
No witness on nulls	Null; Adult	0/2 400; 0/1 500		confirmed	D	C2, H6
Proj. Δ-CE ≈ Grad-diff	Shortcut, B	−0.30 [−0.33, −0.28]†		not conf.	R	C3
Grad-diff < 0.2, offsets	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$	confirmed	R	H5a
Grad-diff rises with α	Grid	+0.57 [0.53, 0.61]	$4{\cdot}10^{-4}$	confirmed	R	H5b
Grad-diff offset-free	Grid	−0.098 [−0.143, −0.053]†		not conf.	R	H5c
Proj. Δ-CE ≈ Grad-diff	Grid	−0.35 [−0.38, −0.33]†		not conf.	R	H5d
CF-diff (no threshold) flags $\lvert b\rvert < 2\kappa$	Logit offsets	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$	confirmed	D	H7c
Witness normal narrows gap $\geq \frac{1}{2}$	Grid	$-0.19 \to +0.03$ [0.01, 0.05]†	$6{\cdot}10^{-4}$	confirmed	R	H8

4 Results

Findings are grouped by the kind of change; Table 3 lists every test.

4.1 Offsets

Gradient differences are blind to an offset, as Proposition 2(a) predicts: on the pure offsets of the grid their Recovery is 0.081 (below 0.2 in each of the 4 pairs), and on Threshold Δ-CE beats them (0.41 vs. 0.10 on split B; positive for each of the three seeds). On an offset the changed factor is the task factor, so even CF-new recovers it (0.90–0.98). Type-N witnesses exist only on the side the offset lowers (on Threshold: every class-1 query, none of class 0), which is why Δ-CE scores 0.47 in Table 2; type O covers the other side. On split C (three Threshold pairs, 900 queries, descriptive) type N finds a witness for every class-1 query and none of class 0 (Recovery 0.45), type O the converse, and the nearer of the two recovers aspect on all 900 queries (1.00; CF-diff 0.77, CF-new 0.89, Grad-diff 0.07). For an exact logit offset b, witnesses appear only once $|b|$ reaches 2κ and then, empirically, on every query of the lowered class. CF-diff, with no threshold, flags 0.961 of queries already at $|b| \in \{0.5, 0.9\}$, where 3–8% of decisions do flip: that change is real and the witness misses it by design (Corollary 1), a difference in sensitivity, not an error of CF-diff.

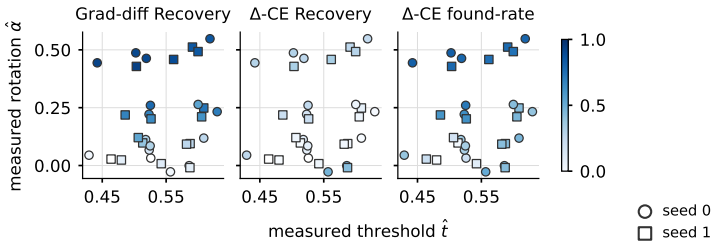


Fig. 1: Grid results at the measured rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} (test split), on one colour scale from 0 to 1. Hollow: no $k^\star$; marker shape: seed of f_{old}.

4.2 Rotations

Gradient differences see a rotation, increasingly with its angle: Grad-diff Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ (each $\alpha = 0.5$ pair above each $\alpha = 0.1$ pair; Fig. 1), while CF-new recovers colour on ≤ 0.03 of points. Proposition 2(b) predicts that after projecting out $\hat{w}_o$ witnesses and gradient differences coincide up to scale; with $\hat{w}_o$ taken at the query they do not (Shortcut 0.64 vs. 0.95 on split B, all valid points; grid -0.35). A post-hoc explanation, tested on new data (four grid pairs, points with a witness): with $\hat{w}_o$ taken at the witness the projected gap Δ-CE $-$ Grad-diff moves from -0.190 to $+0.029$ (narrowing by more than half in each of the 4 pairs, by a small margin in two). This locates the failure, not its cause. On Shortcut (points with a witness) it moves from -0.166 to -0.024; the normal turns by a median 32° from z_0 to $z^\star$, not more for distant witnesses as registered. The witness direction itself follows the analysis: where the condition of Proposition 2(b), computed from margins and gradients at z_0, holds, witnesses recover colour on 0.64 of points, elsewhere on 0.04 (24 rotation pairs, new data, 3 952 witnesses; gap positive in each of the 21 pairs with both groups; descriptive).

4.3 Null updates and abstention

All three disagreement searches verify the margin, and none returns a point on a null update (Table 2). Δ-CE adds proximity: No-prox finds witnesses on fewer points (0.40–0.63 vs. 0.47–0.84) and, where both find one, farther away on every point (median 0.31–0.33 vs. 0.20–0.23). On the 8 null pairs no witness exists, while CF-diff attributes a change to 99.1–99.8% of points. Part of this contrast is the threshold. Scoring a query by $\|u_n - u_o\|$ and cutting on the 2 400 null queries so that none is flagged, as for Δ-CE, CF-diff flags at most 0.001 of the logit-offset queries up to $|b| = 2$, while Δ-CE has a witness for every lowered-class query from $|b| = 1.1$. For a constant offset Corollary 1 makes the witness an exact test of $|b| \geq 2\kappa$, so we repeated this on fine-tuned offsets, where it is not (three Threshold pairs of split C, four $\alpha = 0$ grid pairs): at the same cut CF-diff flags no lowered-class query in any pair, and only 0.05–0.28 at 5% false positives, while Δ-CE has a witness for every one in 6 of 7 pairs and 0.25 in the

seventh (descriptive, saved outputs). On five null updates of an MLP on Adult, no witness exists on 1 500 test points (CF-diff 97.7–100%). NOT_FOUND is no solver failure: 10^5 samples per query find none on 550 null or impossible-class queries.

5 Discussion and limitations

Witnesses are meant to show *where* decisions changed near a query. We have partial checks, not a score: each witness is verified (by definition), the search misses few (in the sampling audit, 3 of 395 small-offset queries) and returns nearer witnesses than No-prox; other families return directions, so this is no comparison. **Limitations.** (1) The image data are synthetic; the tabular test covers abstention only; of label shift, only the prior correction [13] was tested. (2) κ is an exact threshold only for a constant offset; otherwise abstention tracks decisive change (at shifts of 0.03–0.04 it depended on the fine-tuning recipe). (3) Witnesses see only decision changes on the manifold within the budget, with no certificate for deep networks. (4) Recovery scores one notion of change, the factor along which the weights moved. (5) Several tests were formed after earlier results, one after a validation run showed the effect. Every protocol change was logged, with dated commits, before the data it concerns were generated (the analysis-code correction excepted); log, proofs, code: [repository link].

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